

ARJUN BINDU JAYACHANDRAN

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Senior Software Engineer building GenAI and backend platforms in regulated fintech. Initiated a two-tower + LLM-agent bond recommendation platform now in productization; designed LangGraph Orchestrator + Validator RAG systems with enterprise AI governance; hardened PnL microservices with Redis-externalized state, MongoDB aggregation, Kafka exactly-once semantics, and idempotent ingestion under high-volume trade load. Recipient of Wells Fargo's Top Achievers Award 2025.

EDUCATION

National Institute of Technology Calicut

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.55

2022

Calicut, India

TECHNICAL SKILLS

Languages: Python (primary), Java

AI/ML & Agentic: LLMs, Retrieval-Augmented Generation (RAG), LangGraph, LangChain, LLM Agent Patterns (Orchestrator + Validator), Model Context Protocol (MCP), Tool Calling, Vector Search (FAISS), Embedding Pipelines, Two-Tower Neural Retrieval, MMR Retrieval, InfoNCE Contrastive Learning, Evaluation Frameworks (Recall@K, MRR, nDCG), PyTorch, ONNX-Runtime, PII Masking, NoSQL Injection Guards, Tamper-Evident Audit Logging

Backend & Distributed Systems: System Design, Microservices, Event-Driven Architecture, Apache Kafka, REST APIs, gRPC, MongoDB (Aggregation Pipelines, Compound Indexing), Redis (Distributed Caching), SQL, Spring Boot, Apache Camel, Horizontal Scaling, Autoscaling, Idempotency, Server-Sent Events

Financial Domain: Fixed-Income Trading, PnL Microservices, Trade Reconciliation, Swap Position Roll-ups, DMN Rules Engine, Asset-Backed Finance (ABF) Ingestion

Infrastructure, Observability & CI/CD: Kubernetes, Docker, Helm, OpenShift (OCP), GitHub Actions, Jenkins, Prometheus, Grafana, Splunk, Sonar

EXPERIENCE

Wells Fargo | *Senior Software Engineer* | Bengaluru, India

Jan 2026 – Present

- **Initiated a Bond Recommendation Engine** for fixed-income sales-analyst augmentation – built a two-tower dual-encoder PoC (symmetric **InfoNCE contrastive** learning, ANN bond-similarity search) with an **LLM agent layer** that converses with traders and orchestrates similarity search and free-form queries; drove from PoC to a funded production build and now architecting the deployment with **decoupled training + serving** (ONNX-runtime inference, **FAISS** vector indices, temporal train/val/test split) and feedback-driven **routine (scheduled) retraining** from trader accept/dismiss signals.
- **Shipped a replay-safe historical rerun API for PnL Service** – async Camel endpoint with context-propagation headers (recalc-flag, snapshot-date) flowing through trade / position / publish routes and **date-correct MongoDB persistence with Kafka-publication suppression for backfills**, so corrected snapshots never leak into live downstream consumers.
- **Drove reliability hardening across PnL reporting services** – led root-cause analysis on data mismatches, decoupled fragile upstream dependencies, and added fail-fast integrity guards – shifting services from patch-driven fixes to a **resilient-by-design** posture.

Wells Fargo | *Software Engineer* | Bengaluru, India

Jul 2022 – Dec 2025

- **Designed a LangGraph Orchestrator + Validator agentic pipeline** for natural-language → MongoDB query translation – backed by an **MMR-RAG retrieval layer** (dual vector stores for schema + user-context injection), an **MCP server** exposing DB-read primitives to the LLM, and a **governance layer** (NoSQL injection guards, MQL syntax checks, PII masking, tamper-evident audit logging) enforcing safe-query policy before any DB execution.
- **Enabled horizontal scaling** of PnL microservices – externalized in-process state to Redis to enable stateless replicas, with **Kafka exactly-once consumer semantics** ensuring concurrency safety under high-volume trade load; contributed a **DMN rules engine** for swap position and payment-leg roll-ups and idempotent file ingestion for end-of-day archival flows.
- **Optimized MongoDB query latency by 30–40%** on PnL reporting stores – replaced legacy Views with aggregation pipelines and compound indexing, materially reducing infra load across the reporting cluster.
- **Redesigned a PnL data-collection backup microservice from single-call to batched processing** – consolidated per-record upstream calls into bulk payloads, improving end-of-day throughput by **~90%** and reducing I/O contention across the service mesh.
- Authored a **150+ financial-scenario automation framework** (parallel execution, **60% regression-cycle cut**, Splunk telemetry); led **OpenShift modernization of 10+ core services** with resilient CI/CD; owned PnL pricing services and ABF data ingestion with zero production data-quality incidents across the ownership window.

Wells Fargo | *Technology Intern* | Bengaluru, India

May 2021 – Jul 2021

Built a time-series forecasting engine for ATM cash optimization benchmarking **SARIMA, LSTM, and Prophet** on transaction data from 111 ATMs, incorporating exogenous factors (holidays, weather) and automated retraining; achieved **~91% weekly prediction accuracy** via aggregated-window forecasting.

PUBLICATIONS & AWARDS

Publication: *SaBMOGWO-S* – novel algorithm for feature selection across 21 benchmark datasets; The Journal of Supercomputing, 2023 (Vol. 79, pp. 12869–12901), co-authored.

Awards: **Top Achievers Award 2025** (Wells Fargo's highest individual honor); **Innovation Pitch Day 2024 Grand Finalist** (top 4 of 600+ firm-wide submissions); **Manager Spotlight Awards (twice)**.